

The Neural Network

Unit 2

MDS 655

Building Intelligent Machines

The Brain is, inherently, what makes us intelligent

Incredible organ

- stores memories
- experience emotions
- Dream

Building Intelligent Machines (continued...)

Powerful than supercomputers

- Face recognition
- Discern discrete objects from their backgrounds
- Can understand grammar
- Enhancement of vocabulary

Building Intelligent Machines (continued...)

Intelligent machines are build by using brains

- Robots
- Disease detection
- Automated vehicles

Building Intelligent Machines (continued...)

Building Intelligent machines is a challenge

- Needs to solve Computational problems
- Developing radically different way of programming

This leads to deep learning

The Limits of Traditional Computer Programs

Computer Programs are good only at

- performing arithmetic really fast
- explicitly follows a list of instructions

The Limits of Traditional Computer Programs

Brain can recognize every digits though are drastically different

Programs find it difficult specially when the character is not complete



The Limits of Traditional Computer Programs (Continued...)

zero that's algorithmically difficult to distinguish from a six

How to handle this situation?

- Threshold level
- cutoff value of distance to form a close loop



Infinite rules for every situation is not possible

The Limits of Traditional Computer Programs (Continued...)

- We don't know what program to write because we don't know how it's done by our brains
- even if we did know how to do it, the program might be horrendously complicated.

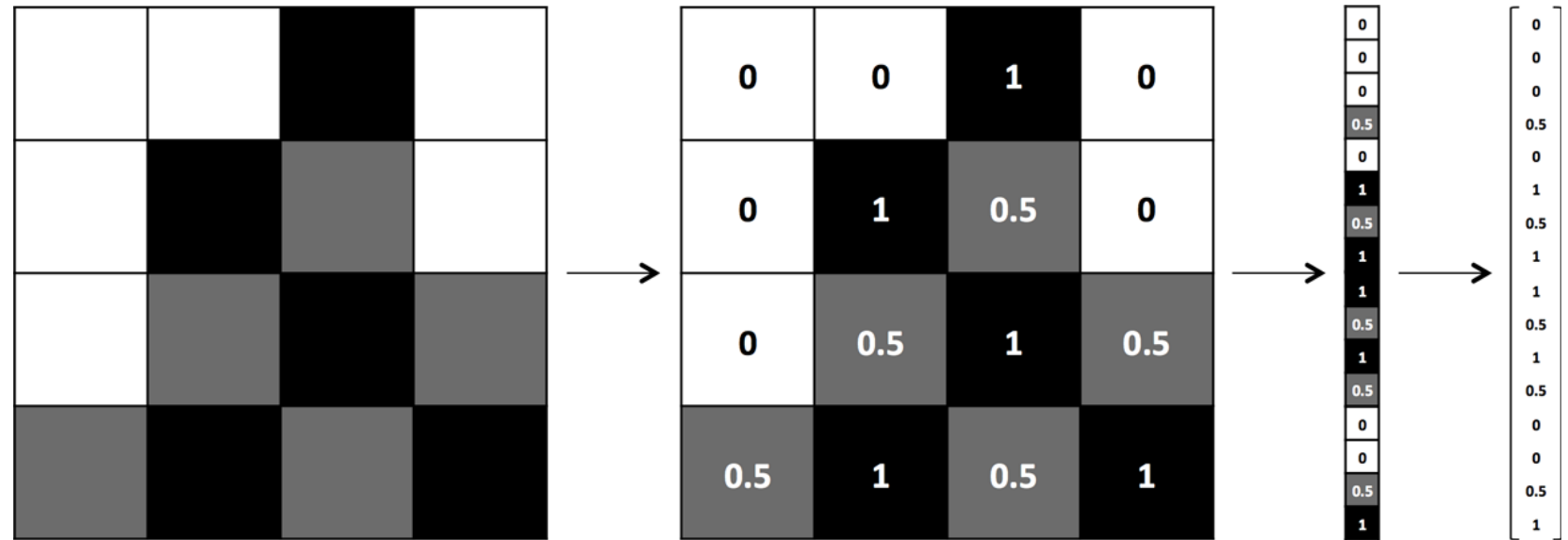
The Mechanics of Machine Learning

$$h(x, \theta)$$

X: vector

(pixel intensities
of gray scale image)

Θ : Vector of parameters that model uses



The Mechanics of Machine Learning (continued...)

$$\mathbf{x} = [x_1 \ x_2]^T$$

x1: number of hours of sleep
x2 : number of hours of study

$$\theta = [\theta_0 \ \theta_1 \ \theta_2]^T$$

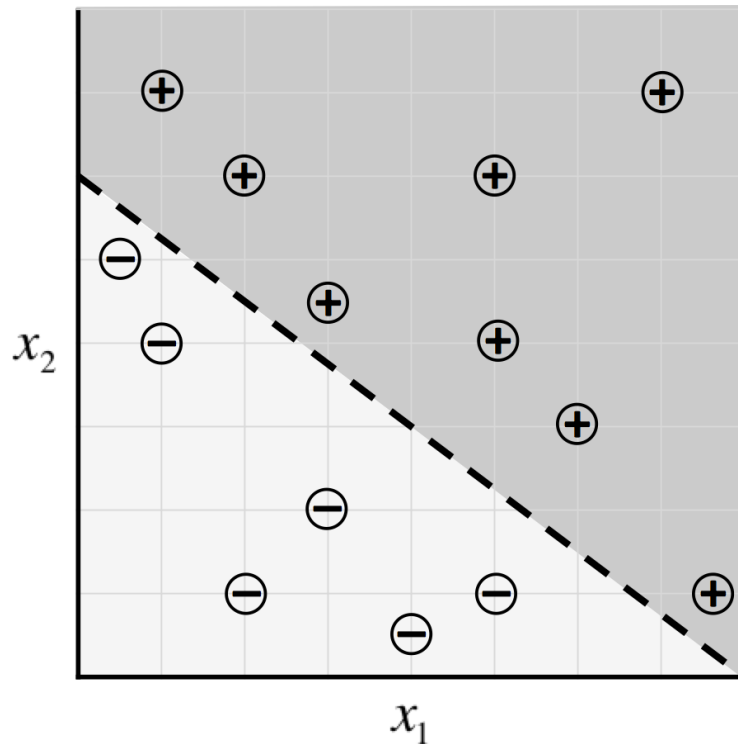
Parameters that is to be learned

$$h(\mathbf{x}, \theta) = \begin{cases} -1 & \text{if } \mathbf{x}^T \cdot \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + \theta_0 < 0 \\ 1 & \text{if } \mathbf{x}^T \cdot \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + \theta_0 \geq 0 \end{cases}$$

The Mechanics of Machine Learning (continued..)

$$\theta = [-24 \ 3 \ 4]^T$$

$$h(\mathbf{x}, \theta) = \begin{cases} -1 & \text{if } 3x_1 + 4x_2 - 24 < 0 \\ 1 & \text{if } 3x_1 + 4x_2 - 24 \geq 0 \end{cases}$$



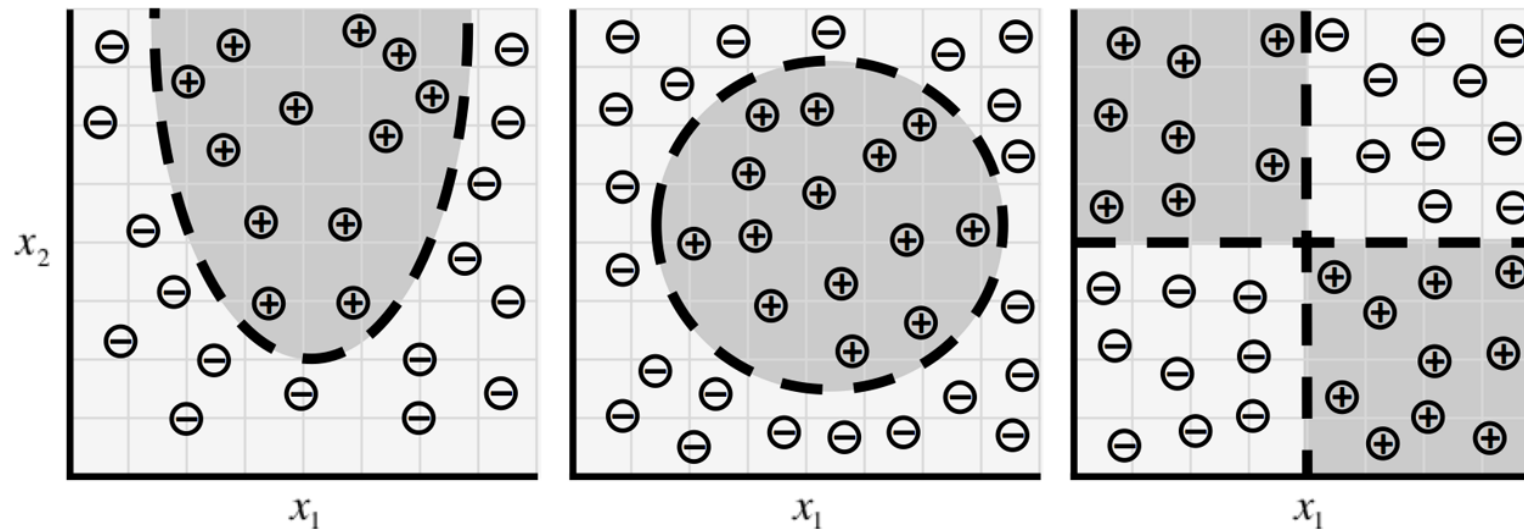
- Geometrically, this particular blueprint describes a linear classifier that divides the coordinate plane into two halves
- This model is called Linear Perceptron

The Mechanics of Machine Learning (continued..)

- Optimization of parameter θ is required
- An optimal parameter vector θ positions the classifier correctly with less error
- This is related to gradient descent

The Mechanics of Machine Learning (continued..)

- As our data takes on more complex forms, we need more complex models to describe them

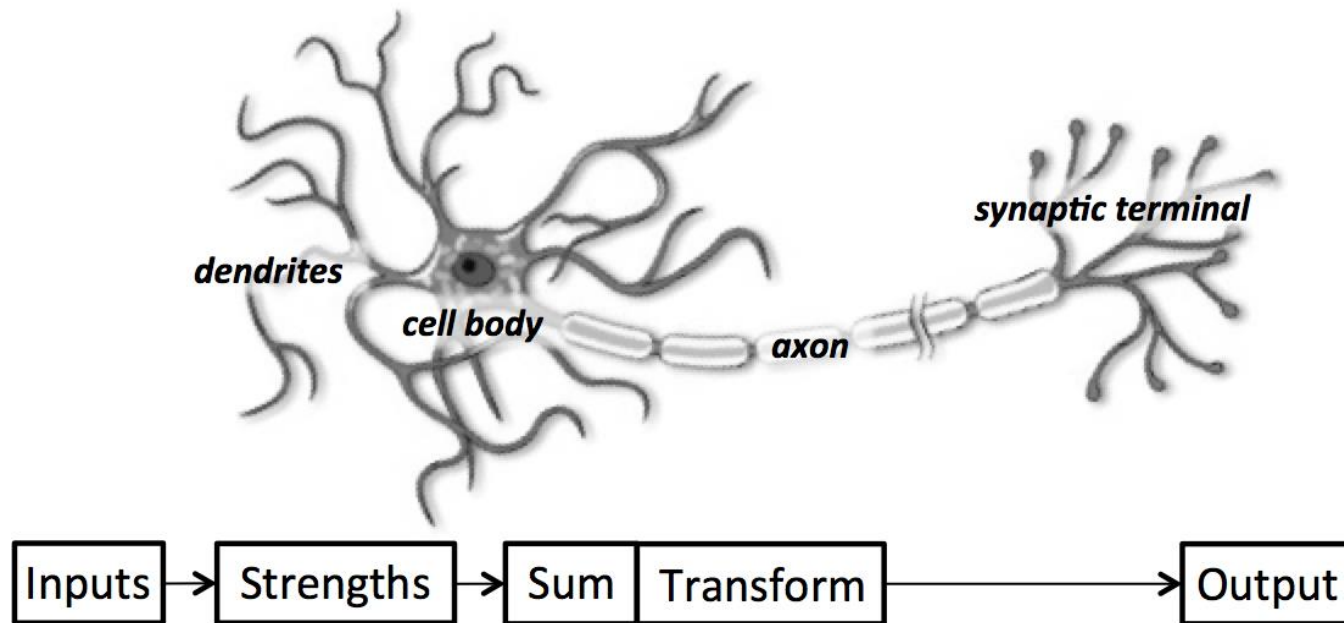


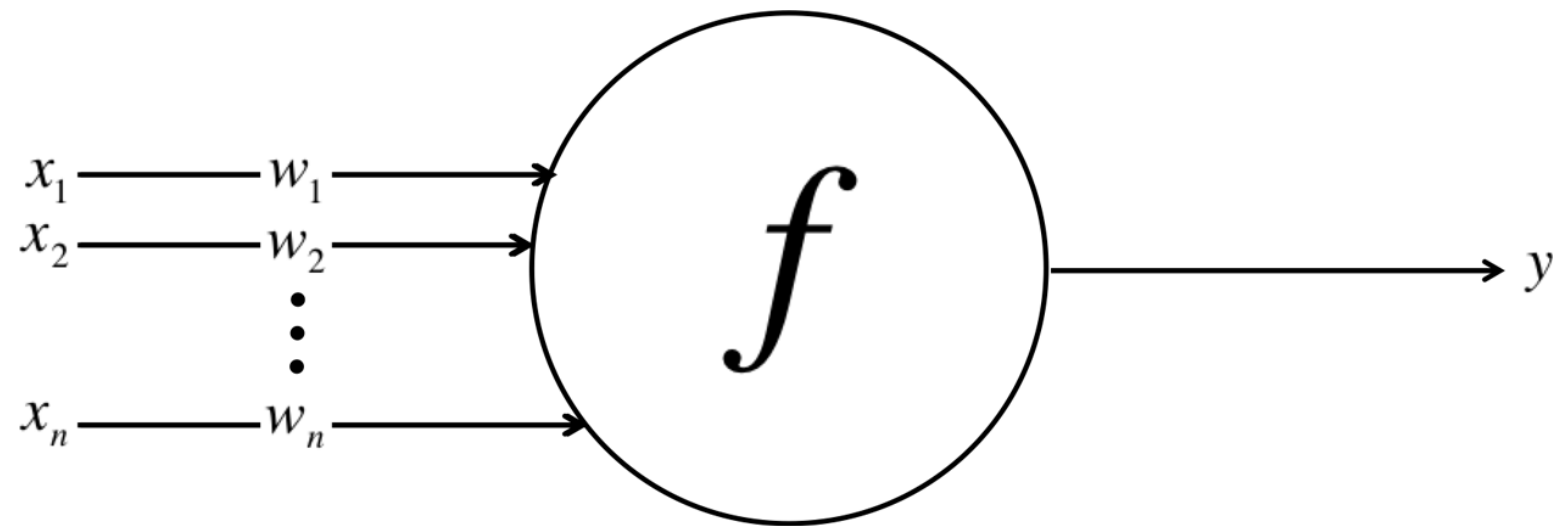
The Mechanics of Machine Learning (continued..)

- Objects recognition, text analysis, image analysis, NLP, computer vision are complex problems
- Extremely high dimensional
- Highly non linear

Requires deep learning i.e. complex models to solve them !!!

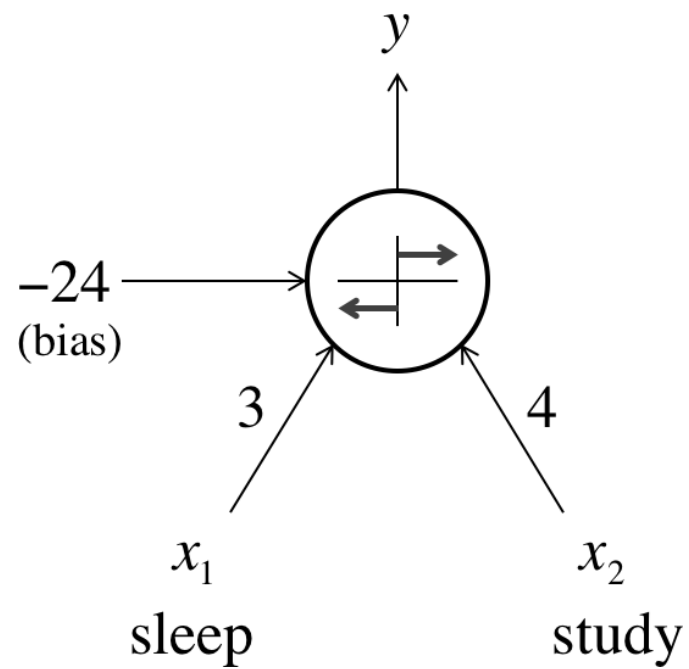
The Neuron





$$h(\mathbf{x}, \theta) = \begin{cases} -1 & \text{if } 3x_1 + 4x_2 - 24 < 0 \\ 1 & \text{if } 3x_1 + 4x_2 - 24 \geq 0 \end{cases}$$

$$f(z) = \begin{cases} -1 & \text{if } z < 0 \\ 1 & \text{if } z \geq 0 \end{cases}$$



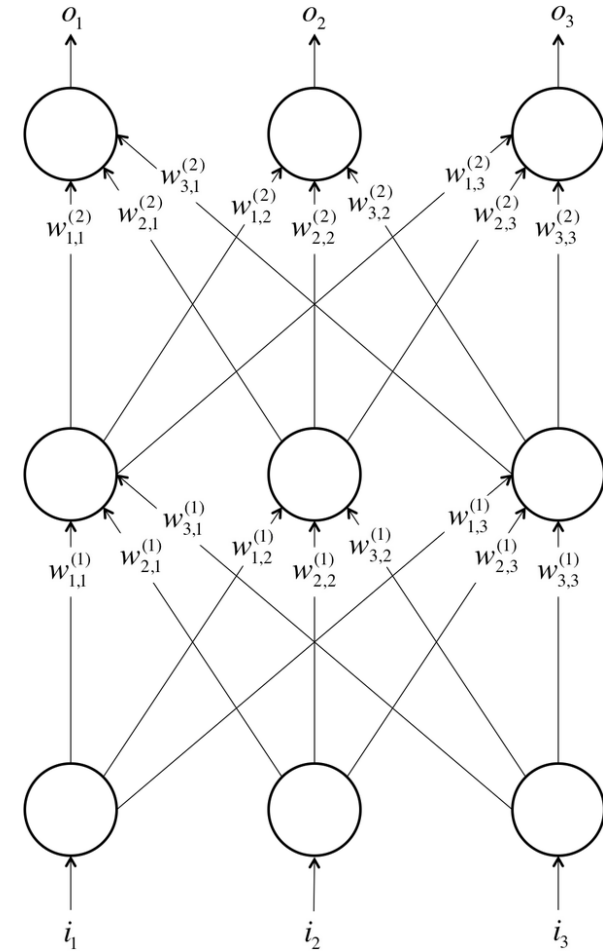
Feed Forward Networks

Information flows from

- one layer to another
- until sensory input is converted into conceptual understanding.

Analogy to Brain

- Bottom most layer visual inputs
- The information flows to higher level
- Sixth layer decides which object



Feed Forward Networks (continued...)

As we mentioned, the layers of neurons that lie sandwiched between the first layer of neurons (input layer) and the last layer of neurons (output layer) are called the hidden layers. This is where most of the magic is happening when the neural net tries to solve problems

Feed Forward Networks (continued...)

Although in this example every layer has the same number of neurons, this is neither necessary nor recommended. More often than not, hidden layers have fewer neurons than the input layer to force the network to learn compressed representations of the original input

Feed Forward Networks (continued...)

It is not required that every neuron has its output connected to the inputs of all neurons in the next layer. In fact, selecting which neurons to connect to which other neurons in the next layer is an art that comes from experience

Feed Forward Networks (continued...)

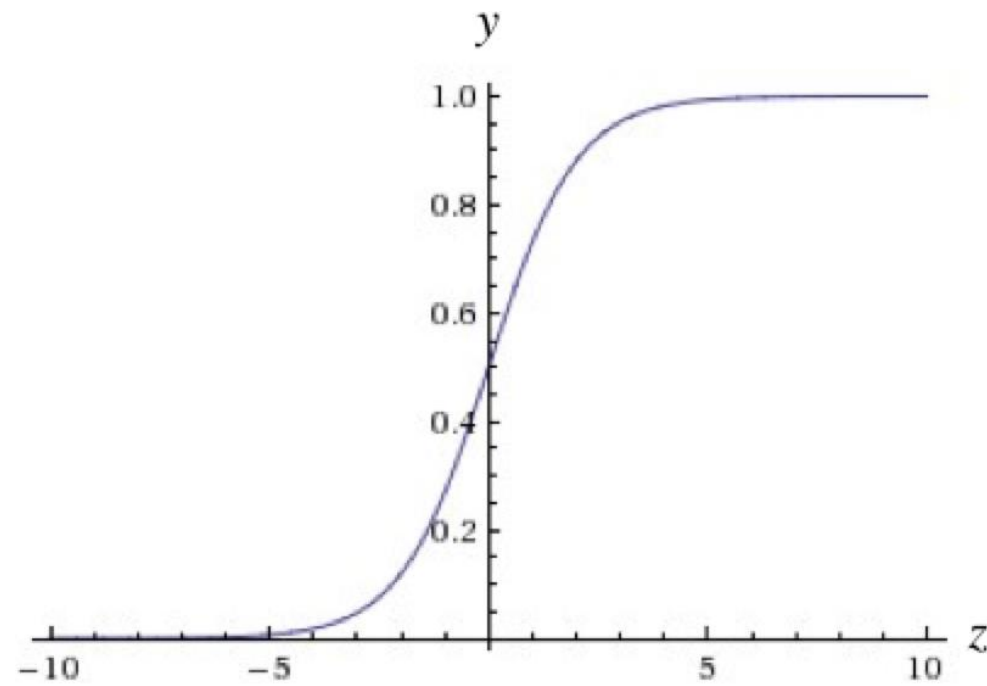
The inputs and outputs are vectorized representations

$$\mathbf{x} = [x_1 \ x_2 \ \dots \ x_n]$$

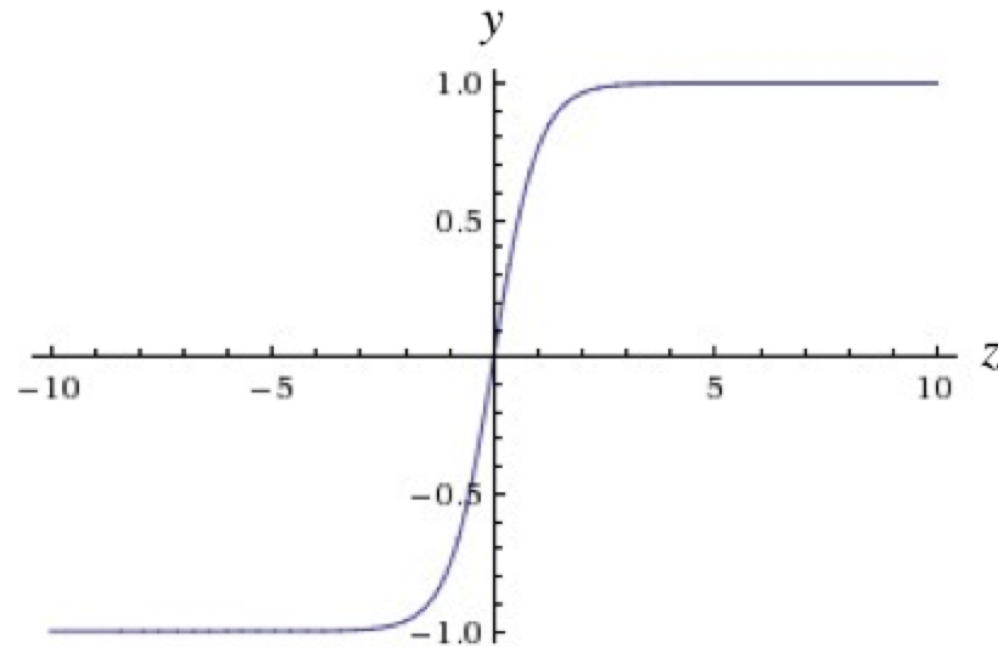
$$\mathbf{y} = [y_1 \ y_2 \ \dots \ y_m]$$

$$\mathbf{y} = f(\mathbf{W}^T \mathbf{x} + \mathbf{b})$$

Activation Function



Activation Function (continued...)



Activation Function (continued...)

